Akshaya Tharun C R

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.02
Maharishi International Residential School,Sriperumbudur	Class XII (CBSE)	2023	97.40%
Senthil Public School,Salem	Class X (CBSE)	2021	96.00%

SCHOLASTIC ACHIEVEMENTS

NTSE scholar (A scholarship provided to students based on a competitive examination testing aptitude , math and science).

ICPC Asia West Regionalist - Ranked 22nd in ICPC Asia West 2025-2026.

POSITIONS OF RESPONSIBILITY

Held the position of **Core** Member in Infero(Algorithmic and Logical Reasoning Club of IITH) .

Conducted an introductory session on **graph theory** useful for competitive programming.

WORK EXPERIENCE

Frontend Performance Analyzer -- AI-Driven Development

- Designed a **Page Performance Analyzer** entirely through Cursor commands by reasoning about frontend component interactions and identifying high-value diagnostics for rapid engineering debugging.
- Built an **end-to-end autonomous workflow** that navigated the codebase, performed component-level analysis, generated the required implementation, and iteratively validated the solution without manual coding.

RELEVANT COURSES

Computer Science : Discrete Mathematics for Computer Science , Software Development Fundamentals

Data Structures and Applications , Computer Architecture , Operating Systems , Database Management System , Theory of Computation , Algorithms , Compilers - 1 , Introduction to Programming .

Mathematics : Calculus , Probability and Random Variables , Linear Algebra , Transform Techniques , Intro to Statistics , Convex Optimization .

SKILLS

- **Programming Languages -** C , C++ , python , HTML , MySQL , RISC - V Assembly .
- **Problem Solving -** Max Rating of 1712 in Codeforces([profile]
- **Utilities -** LaTex , Linux , Git .

PROJECTS

Mini Project -- Streaming Algorithms for Graphs

- Studied the problem of finding long induced paths in K_{s,s}-free graphs, developing and analyzing **graph-theoretic and streaming-algorithmic techniques** under two faculty advisors.

VortexTV - Full-Stack Movie Streaming Web App

Developed a movie streaming platform with features like watchlists, history tracking, and personalized recommendations. Built the frontend using **React.js (Vite) & TailwindCSS** and the backend with **Node.js, Express, and MySQL**, implementing secure user authentication and content management **[GitHub]** .

Virtual Memory Management in gemOS (OS Kernel Project)

Implemented core memory management features including mmap, munmap, mprotect, page fault handling, and a Copy-on-Write (cfork) fork system call in the gemOS educational kernel.

Handled dynamic VM area allocation, page table manipulations, and TLB-consistent access control using custom kernel code in C **[GitHub]**

RISC V Simulator

- Implemented a command line **RISC V assembler** using C++ to generate machine code(hex code) from RV32I assembly code , handling full instruction formats (R, I, S, B, U, J) with robust error detection.
- Built an **interactive simulator** supporting memory/register modeling, step-by-step execution, breakpoints, call stack tracing, and command-line controls (e.g., run, step, mem, regs, show-stack).
- Conducted extensive **cache performance analysis** using the Ripes simulator, exploring impacts of block size, associativity, and write policies on hit/miss behavior.

EXTRACURRICULAR

- Represented **IIT Hyderabad** as a member of the **Basketball team** in the **Inter IIT Sports Meet 2024**.
- Won 2nd place in Institute **Water Polo** League .

Interests: Avid reader of various genres(fiction & non-fiction) such as cognitive psychology , philosophy , motivational , self development , thriller , mystery etc ...